Zooming Into Video Conferencing Privacy

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***Abstract*— The unprecedented growth in video conferencing usage is accompanied by multiple security and privacy threats. Importantly, protecting users' privacy is not always in their own hands. Posting meeting images affects all participants, leading to an easy collection of personal data including age, gender and linkage with participation in other meetings. Here, we explored privacy issues that may be at risk by attending virtual meetings. We extracted private information from collage images of meeting participants that are publicly posted online. We used image processing, text recognition tools, as well as social network analysis to explore our curated dataset of over 15 700 collage images, and over 142 000 face images of meeting participants. We demonstrate that video conference users are facing prevalent security and privacy threats. Our results indicate that it is relatively easy to collect thousands of publicly available images of video conference meetings and extract personal information about the participants, including their face images, age, gender, usernames, and even full names. This type of data can vastly and easily jeopardize people's security and privacy both in the online and real-world, affecting not only adults but also more vulnerable segments of society, such as children and older adults. Finally, we show that cross-referencing facial image data with social network data may put participants at additional privacy risks they may not be aware of and that it is possible to identify users that appear in several video conference meetings, thus providing a potential to maliciously aggregate different sources of information about a target individual.**

***Index Terms*— Data science, image processing, security and privacy, video conference, video conference applications.**

I. Introduction

THE COVID-19 pandemic outbreak has had tremendous impacts on daily life in multiple aspects, ranging from basic hygiene, medicine, health science, economics, politics, and society. In particular, it dramatically affected how people communicate with their families, friends, and coworkers [1]. Social distancing resulted in the search for novel ways to collaborate, study, celebrate birthdays, and even meet with parents and grandparents. One of the most popular emerging solutions are using video conferencing applications, such as Google Meet,[1] Microsoft Teams,[2] and Zoom.[3] During the COVID-19 outbreak and accompanying stay-home quarantine strategies, people have started using video conferencing applications to replace face-to-face meetings in schools, workplaces, and social gatherings with virtual meetings. As a result, the number of video conferencing users along with the number of daily meetings surged sharply [2]. Today, there are hundreds of millions of video conferencing meetings that take place daily, encompassing millions of users [2].

With the unprecedented growth in video conferencing usage, many security, and privacy issues have been unraveled [3], [4], [5], [6], [7]. These issues range from unencrypted communication for unpaid users [3] to vulnerabilities that allow malware execution on participants' devices [4]. Moreover, the use of video conferencing applications has raised privacy concerns that may enable uninvited attendees to find ways to join meetings by guessing meetings IDs [5] or by simply searching for video conferencing meeting links that were made publicly available (e.g., meeting links published on social media websites [6]).

A malicious user that gains access to video conferencing meetings can collect sensitive and private data on users [8]. This includes their names, usernames, images of their faces, samples of their voice, and even exposure to personal data that has been shared as part of the conversations. Moreover, using accessible deepfake tools, a malicious user can attend video conferencing meetings under a false identity. For example, one can use realtime deepfake tools in order to join meetings using celebrity avatars. As a result, he or she will look like celebrities, such as Barack Obama or Elon Musk [9].[4] The participant's personal data can later be exploited to jeopardize participants' safety in both the virtual and the real world [10]. For example, Acquisti et al. [11] demonstrated how to face recognition technology could be used to identify individuals both in the online and offline worlds. Namely, they used publicly available images from Facebook to identify students strolling through campus. They also illustrated that it is possible to predict personal and sensitive information from a face, such as the individual's interests, activities, and even his or her social security number.

In this study, to the best of our knowledge, we provide the first in-depth look into the world of video conferencing by analyzing images collected from thousands of video meetings publicly published online. Unlike regular images on social media, video conference images have a predefined structure and participant names embedded in them. Additionally, when taking regular group photographs, the people in the picture know the photograph was taken. However, in online meetings, participants do not necessarily know a photograph was taken, and standard video conferencing security mechanisms like passwords are not practical here. These factors make video conference images more useful to attackers. Using the additional data, an attacker can perform attacks that will be almost

Manuscript received 27 July 2022; revised 30 October 2022 and 9 December 2022; accepted 20 December 2022. Date of publication 2 January 2023; date of current version 31 January 2024. *(Corresponding author: Dima Kagan.)*

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Digital Object Identifier 10.1109/TCSS.2022.3231987

[1] https://meet.google.com

[2] https://www.microsoft.com/en-us/microsoft-365/microsoft-teams/group-chat-software

[3] https://zoom.us

[4] https://github.com/alievk/avatarify

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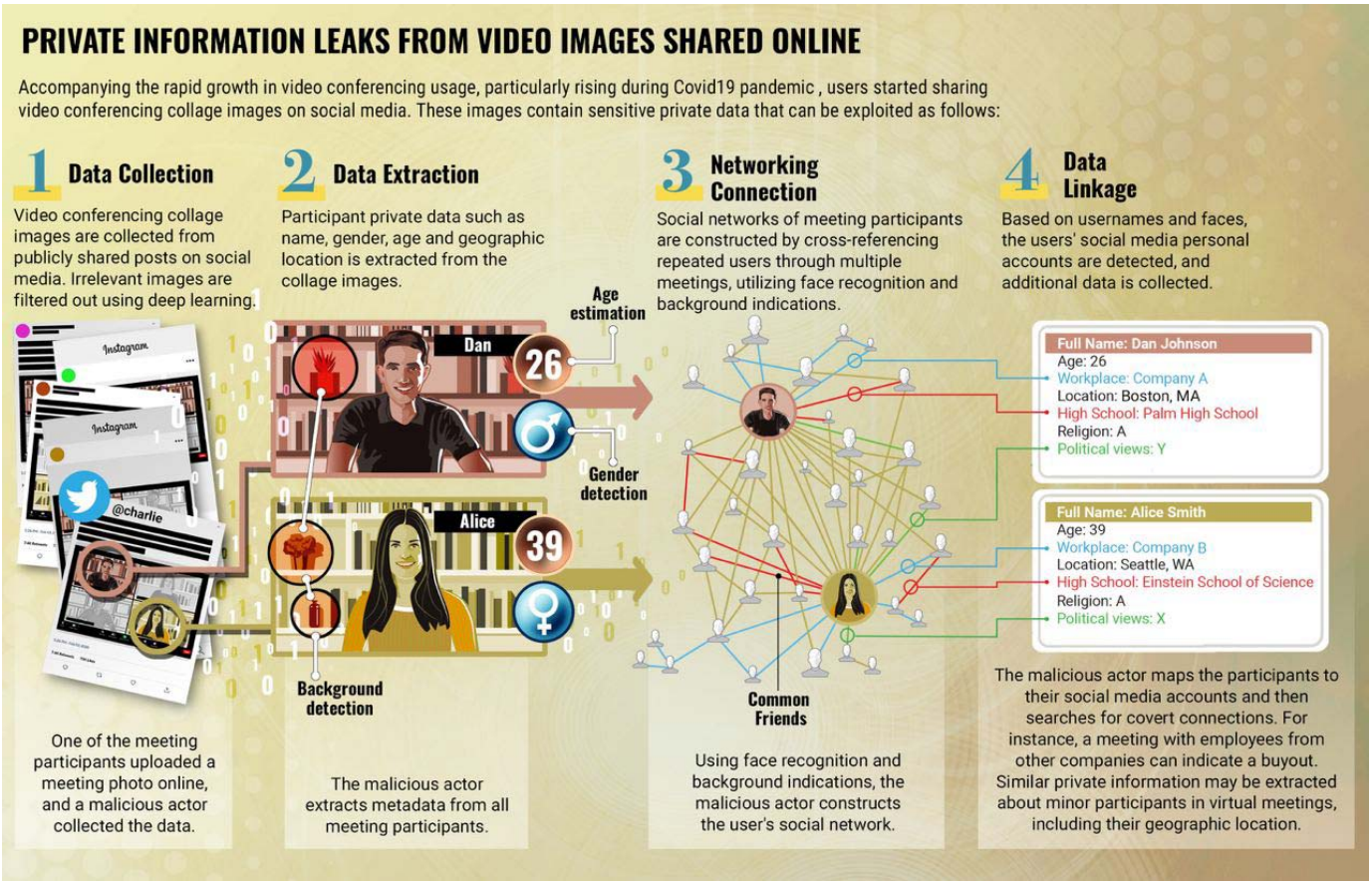

Fig. 1. Private information leaks from video images shared online infographic.

impossible with regular online images. We curated an image dataset by collecting and analyzing ten-of-thousand images published on social media websites. We subsequently analyzed the dataset to extract information from each image regarding the meeting participants, such as their age, gender, username, as well as several other features.

We compute statistics on video conferencing usage and demonstrate that video conferencing participants encompass a wide range of ages, from young children to older adults (see Fig. 1). We apply deep-learning-based image processing algorithms to demonstrate that it is possible to identify the same individual's participation at different meetings by simply using either face recognition or other extracted user features. The extracted information about users has the potential to be harmfully used to uncover participants' social networks and other privacy-related factors (see Fig. 2). Overall, our study emphasizes the privacy risks that video conferencing participants need to be made aware of. We demonstrate that there is danger not only from a malicious user but also from other users in the meeting that can innocently upload *a single photograph*, which can be maliciously used to affect their and their family's privacy.

In this study, we explored privacy issues in the world of video conferencing by analyzing images collected from thousands of video meetings (also illustrated in Fig. 1), which were publicly published online on social media websites. We demonstrated that it is relatively easy to extract private information about meeting participants, even in cases where they themselves have not posted online. Based on the extracted data, we constructed a social network of the meeting participants revealing hidden social ties. Additionally, we demonstrated that even the user background could be used to connect between different meetings and people. The disturbing risks apply to all age populations, including vulnerable populations such as young children. We emphasize how video conferencing makes it even more difficult to deal with multiparty privacy (MP) issues in social networks.

II. Background

Over the years, many efforts have been put into analyzing user security and privacy and offer solutions in various online platforms. These efforts include online trading systems [12], online social networks [10], cryptocurrency, platforms [13], [14] and Internet of Things (IoT) [15]. In this study, we focus on privacy and security issues in video conferencing applications, one of the most commonly used ones known as Zoom. Such applications have rapidly emerged as cardinal ways of communication as a result of stay-home and social distancing orders due to the COVID-19 pandemic outbreak. Video conferencing applications have been supporting, in addition to work-related and educational activities, also a wide range of social activities, including virtual happy hours, virtual yoga, blind dating, and worship services [16]. Consequently, video conferencing platform users have been exposed to multiple security and privacy risks, similar to those related to online social networks.

Privacy risks related to online social networks include the following.

1) *Information Leakage:* Information leakage refers to the detection and extraction of unintentionally disclosed

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information [17]. Visual data contains implicit information in the background that may reveal interesting, and possibly sensitive pieces of information. For instance, Reece and Danforth [18] demonstrated that it is possible to detect signs of depression in Instagram photographs. Recent studies, such as Hilgefort et al. [19], have demonstrated that it is possible to extract information about the real-world environment even through that virtual background. The virtual background is placed through a computer algorithm that separates the speaker from the room. However, the algorithm sometimes glitches; when it happens, we see some weird artifacts appearing on the screen. Apparently, these artifacts contain information about the physical environment and expose real-world pixels. In doing so, Hilgefort et al. [19] were able to aggregate these pixels to reconstruct enough data to recognize some objects in the real world physical environment, even though a virtual background presumably hid it. Moreover, Yang et al. [20] demonstrated that many video conferencing apps continue to access the microphone regardless of when the application is muted. Some apps even send statistics to their telemetry servers. They proved that based on these statistics, they could predict what kind of activity the muted participants performed. In another study, Wasswa and Serwadda [21] showed that data from the participant's monitor could be leaked through the participant's eyeglasses. Eyeglasses are a reflective surface that mirrors the image on a computer screen. They demonstrated that it is possible to train machine learning to recognize the content reflected in the eyeglasses and even recognize the text. Sabra et al. [22] demonstrated that we leak a lot of data by recording and analyzing body movement while working on the computer. They used the fact that typing on a computer affects upper body movement. To detect keystrokes, they preprocessed the video stream via multiple steps such as removing the background, and extracting the participant's body and face. Then they use a keystroke detection algorithm to measure the interaction between the face, arms, and shoulders. Finally, the extracted data is combined with a dictionary-based prediction algorithm to predict words.

2) *Information Linkage:* Information linkage occurs when an attacker can link several pieces of information from at least two separate data sources. For example, an attacker may match the user's publicly available details in one online account (e.g., username or profile picture) with other accounts and social network profiles of the same user in order to uncover additional details about the user, such as full name, home location, and workplace.

3) *Malware Attacks:* Malware refers to malicious software that is intentionally developed in order to disrupt a computer operation to collect credentials and gain access to private information. Like many other web platforms, video conferencing applications are also subject to malware attacks. For example, in parallel to Zoom's dramatic rise in popularity, it has become a popular target for hackers [4]. In 2020, it was reported that Zoom's zero-day vulnerability allows for remote code execution and malware launching on target computers [4].

4) *Phishing Attacks:* Phishing attacks are a form of social engineering developed in order to steal user-sensitive and private information by masquerading as a trustworthy third party. Since the COVID-19 outbreak, hackers have been increasingly impersonating video conferencing applications, including Zoom, Microsoft Teams, and Google Meet by sending phishing emails and registering various Zoom-related domains for phishing scams [4]. Moreover, a new type of Zoom-related phishing campaign has recently been revealed in which the phishing mail contains a fake Zoom link and includes words like "termination" to distract users from noticing that the link in the email is not linked to Zoom's official website [4].

5) *Face Recognition Attacks:* Face recognition algorithms can identify or verify a person from a digital image or a video source. Identifying a person's face from a video, and cross-referencing it with other datasets might be used to expose personal information about the individual [23]. Acquisti et al. [11] demonstrated that it is possible to identify strangers online (on a dating site where individuals protect their identities by using pseudonyms) and offline (in public space), based on photographs that are publicly available on social network sites.

In addition, users of video conferencing applications are exposed to an additional set of risks as follows.

1) *Data Breach:* Some video conferencing applications provide recording and cloud storage capabilities for their users. In 2020, the Washington Post [4] reported that thousands of private Zoom videos were publicly accessible online on either Zoom's cloud storage, or other external storage services. Washington Post reporters watched several of these videos and discovered that these videos contained sensitive and private information, including people's names and phone numbers, private company and financial statements in small-business meetings, as well as elementary school classes, in which children's faces, voices, and personal details were exposed. Even users that used a password to protect their videos on Zoom cloud were exposed to these same risks. This was demonstrated by a proof of concept example that used brute force to bypass the password-protected videos.[5] Recently, Mahr et al. [24] demonstrated that by utilizing various forensic tools on Zoom user devices, it is possible to extract many pieces of private information; examples include chat messages, profile pictures, exchanged files, user contact information, etc.

2) *Fake Avatars:* Fake avatar is a new type of identity fraud, similar to fake profiles on online social networks [25]. Video conferencing application users can be victims of this new type of identity fraud, in which other participants in a meeting can pretend to be a person who they are not. For example, the code project Avatarify[6] can be used to create photorealistic avatars for

[5]https://github.com/markbuffalo/zoombo

[6]https://github.com/alievk/avatarify

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video-conferencing, which may mislead meeting participants into thinking they are meeting with a celebrity [26]. Similar to other deepfake-based applications, this type of technology might be exploited for fraudulent activities. For example, there are indications that in 2019, fraudsters used deepfake voice software to mimic a CEO's voice in order to steal money from a company [27].

3) *Cyberbullying:* Cyberbullying (also referred to as Cyber Abuse) refers to bullying and harassment that occurs within technological communication platforms, such as chats, mobile devices, and social networks [28]. Recently, users of video conferencing applications have been exposed to a new type of online harassment called *Zoom Bombing*, where an uninvited person joins a video chat meeting and interrupts the meeting by sharing inappropriate content [6].
4) *Zoombombing Campaigns:* Zoombombing (also referred to as Zoom Raiding) is a practice that refers to unwelcome stranger participants joining a meeting, and disrupting video calls with offensive language and imagery. This type of behavior has recently become a dangerous method for coordinated harassment and hate speech [29]. For example, recent research performed by the New York Times uncovered that thousands of people gathered online to organize Zoom harassment campaigns, share meeting passwords and plan for sowing chaos in public and private meetings [29]. Also, the U.K. authorities investigated 120 Zoombombing cases of Zoom video calls that were hijacked by people displaying images of child abuse [30]. A recent study Ling et al. [31] discovered that zoombombings are typically carried out by insiders who had a legitimate invitation to the video conference. They also discovered that these insiders shared private data from the meetings, such as the meeting participants' real identities. While studies have performed video conference threat modeling [32], forensic analyses of security risk in video conference applications [24], and even surveyed users regarding their privacy and security settings [33], to the best of our knowledge no study focused on privacy issues that result from the social nature of communication between people.

III. Methods

The primary goal of this study was to explore privacy aspects that may be at risk by attending virtual conference meetings, a practice that has become very common following the COVID-19 outbreak and related stay-home lockdown and social distancing measures.

We applied the following methods to explore privacy issues that may be at risk by attending virtual conference meetings.

A. Building a Database of Video Conference Images

As a case study, we focused on Zoom, the video conference application whose popularity skyrocketed during the COVID-19 pandemic.[7] We curated an image dataset that contains images from thousands of video conference meetings by performing the following three steps.

First, we created web crawlers that collect data from Twitter and Instagram using online accessible Twitter and Instagram scraper tools.[8,9] We were specifically interested in collecting images that were likely to be taken in video conference meetings. Thus we set the web crawlers to search for tweets that contain terms or hashtags from a predefined set of meeting relevant target terms (e.g., "zoom school" and "#zoommeeting").[10] Using this method we were able to gather 89 305 Zoom-related tweets and 90 395 Zoom-related Instagram posts (a total of 179 700 meeting-related posts).

Second, we filtered out tweets and Instagram posts that did not contain an image and extracted the images from all remaining tweets and posts (a total of 26 408 images from Twitter and 78 435 images from Instagram were collected). We then utilized the Fastai framework [34] to construct a binary image classifier that identifies which of the images contain a Zoom video meeting collage of participants.[11]

To train the classifier, we manually labeled 5271 images, out of which 1505 contained a collage of participants (positive examples) and 3766 were labeled as other types of images (negative examples). Using the labeled images, we applied ResNet-50 [35] based transfer learning to train our model. The trained classifier achieved an accuracy of 0.969, a true-positive rate of 0.935, and a false-positive rate of 0.016 on a test set of 1054 images. This process resulted in a dataset consisting of 16 133 Zoom collage images.

To eliminate duplicates and similar images (i.e., the same image with minor adjustments or crops), we utilized dhash [36] to compute the Hamming distance between all pairs of images. We removed all images with a distance equal to or smaller than an empirically set threshold 1.2.[12] Moreover, to remove additional similar images, we computed the Euclidean and cosine distances between all pairs of images using each image embedding generated by using the Zoom image classifier last layer representation. Then, we removed all for which the cosine and Euclidean distances were equal or lower than 0.0035 and 25 respectively.[13] Finally, after getting rid of duplicate images we were left with an image dataset

[7]The same method with minor adjustments can be applied to almost any other video conference application.

[8]https://developer.twitter.com/en/docs

[9]https://github.com/arc298/instagram-scraper

[10]Our Twitter crawler downloaded tweets that contained the following terms and hashtags: "zoom birthday," "zoom happyhour," "zoom school," "zoom party," "#happyhour," "#Zoom,""#zoommeeting," "#zoomus," and "@Zoom_US." The Instagram crawler downloaded posts containing the following hashtags: "#zoommeeting," "#zoomparty," and "#zoombirthday.' The choice of terms and hashtags was empirically selected to maximize the number of Zoom collages returned by the search engine. For example, using the search term "Zoom" alone is too broad, and mostly returns irrelevant results. In contrast, terms like "#zoommeeting" are unambiguous and had more Zoom video conferencing related posts. We tried to select the most relevant search terms by inspecting the search results for those including Zoom collage images. It is important to note that, unlike Twitter, Instagram does not support free text search, only hashtags. Therefore, we used a large number of hashtags.

[11]The classifier is a binary classifier of input images, with two output classes: either a Zoom collage image or any other image.

[12]The selected threshold of 1.2 was chosen, by manually testing and evaluating several threshold values, searching for minimum false positives.

[13]The selected threshold values of 0.0035 and 25 were empirically chosen, by testing and manually evaluating several threshold values.

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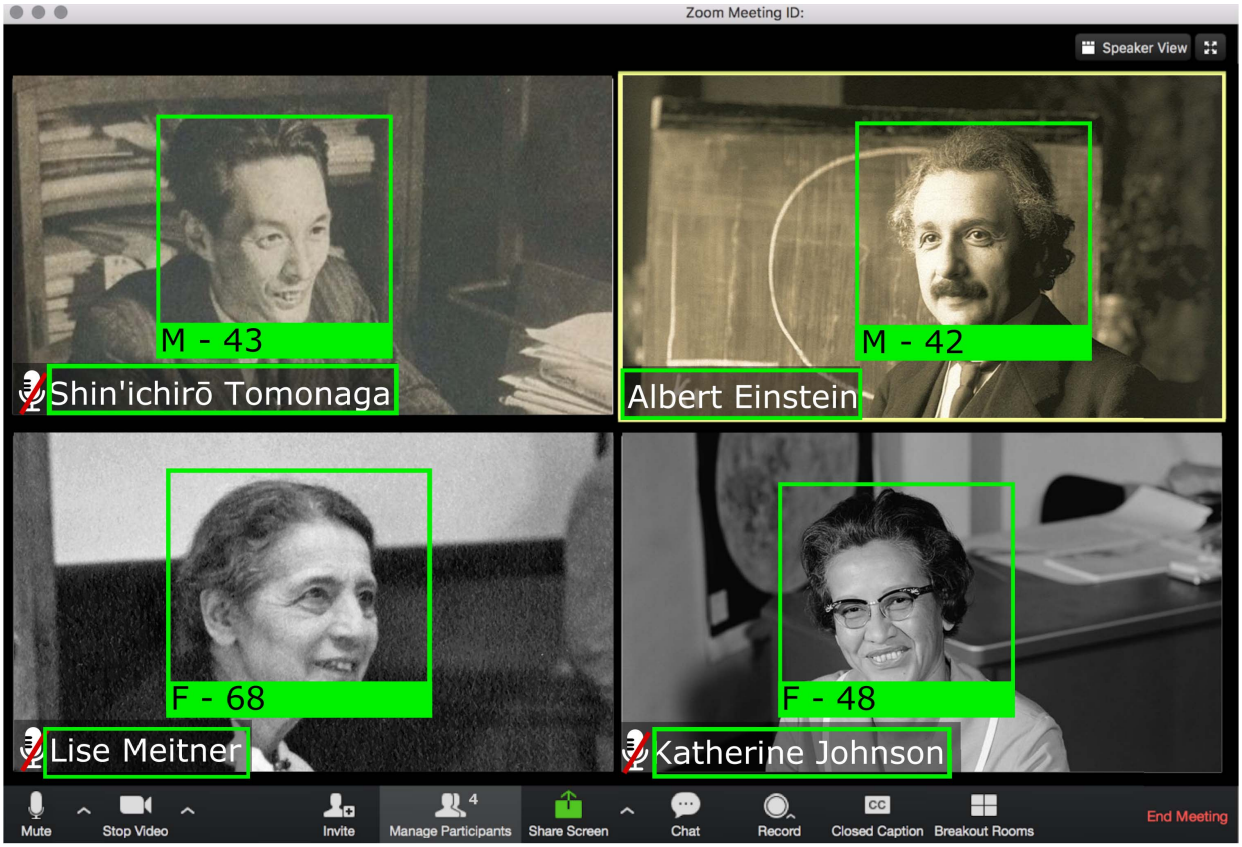

Fig. 2. Zoom image collage with detected information, along with extracted features of gender, age, face, and username.

consisting of 15 709 collage images automatically collected from Twitter and Instagram.

B. Automatically Extracting Private Information

To extract structure data from the collage image dataset, we performed the following steps on each image in the dataset.

1) *Faces Detection:* Zoom collage images typically contain multiple face images. We used two face detection tools to detect each face in the collage. The first tool we used was a pretrained model[14] based on multi-task cascaded convolutional network (MTCNN) [37]. The second tool we utilized was Microsoft Azure Face API for face recognition.[15] Each of the two tools extracts the bounding boxes of the faces in each collage image in our dataset. The results of the two face detection tools are combined as follows: We keep the faces that are uniquely identified by each tool, and in cases where the bounding boxes of two faces intersect, we consider them as the same face (the bounding box from the first model is used). To estimate recall values, we manually evaluated face detection recall rates in 100 randomly selected Zoom collage images. Our face detection approach achieved a recall of 80% (i.e., detection of 80% of faces in the available data).
2) *Face Embedding:* By using a dlib-based tool [38], [39] on each detected face in our dataset, we generated a 128-dimension numerical vector representation of the features extracted from the face, which was later used for creating a face recognition database.
3) *Age Detection:* For each detected face image, we estimated the person's age by applying two separate models: a pretrained model for age detection[16] and the Microsoft Azure Face API package. We define the estimated age as the average of two predicted ages. We categorized the average predicted ages into four categories [40]

$$f(\text{age}) = \begin{cases} \text{child}, & \text{if age} \leq 12 \\ \text{adolescent}, & \text{if age} > 12 \text{ and age} \leq 17 \\ \text{adult}, & \text{if age} > 17 \text{ and age} < 65 \\ \text{older adult}, & \text{otherwise.} \end{cases}$$

4) *Gender Detection:* For each face detected by Microsoft Azure Face API [41], we also utilized the API to determine the user's gender. By manually inspecting 100 randomly selected Zoom collage images, we evaluated that Microsoft Azure Face API detects gender on our curated dataset with a recall of 98.9%.
5) *Username Recognition:* We used a publicly available scene text recognition library to identify participants' usernames.[17] The library implements a text detection approach that is based on EAST [42] for detecting text location and MORAN [43] for subsequently recognizing the composing characters. To filter out words that are not

[14]https://github.com/ipazc/mtcnn
[15]https://azure.microsoft.com/en-us/services/cognitive-services/face/
[16]The code was forked from https://github.com/yu4u/age-gender-estimation and refactored into a library instead of a CLI tool. The modified code is available on https://github.com/Kagandi/age-estimation-pytorchGitHub
[17]https://github.com/gtsoukas/scene_text

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username candidates, we used a list of Zoom meeting-related words.[18] We applied spaCy lemmatization [44] to filter out all dictionary words, aiming at removing words that are not names. One of the issues with the text recognition model is that it detects only single words. However, in many cases usernames are a combination of multiple words, such as first name, middle name, and last name. To combine single detected words into the associated full usernames we applied a custom heuristic approach based on the unique image structure of Zoom collages. We observed that it is rare to find two words located very close to one another that are not a part of the same name. Thus, we heuristically joined recognized words that are closely located in image space into a single username. Namely, for each identified word in each image collage, we searched for another word that is nearest to its top right point. If the nearest word was at a Euclidean distance smaller or equal to 10,[19] we merged both words. We repeated this process to construct usernames of more than two words until no words were combined further. We evaluated this heuristic on 100 randomly selected Zoom collage images and found that it combines 97.6% of the usernames longer than a single word into the correct usernames. Using 100 randomly selected Zoom collage images, we compared the actual username with the detected username to test the username recognition process (scene text detection followed by multiple word merging). The method detected 63.4% of the usernames correctly.[20]

C. Linking Personal User Data to Social Network Data

We explored whether video conference participants are vulnerable to information linkage attacks that utilize involuntary information leakage. An information linkage attack (also called Unwelcome Linkage) "occurs when links on the Internet reveal information about an individual that they had not intended to reveal" [45]. To demonstrate that this type of attack is indeed possible using video conference data, we reconstructed participants' links and social networks by matching users that participated in multiple meetings using the following methods.

1) *Cross Referencing Usernames*: We matched users across video conference meetings by explicitly using participants' usernames. This method considerably reduces the search space for Zoom participant identities. Moreover, in some cases, participants have distinct usernames, which significantly assists in finding the same person in multiple meetings.

2) *Cross Referencing Using Face Embedding*: We matched users across different meetings by comparing the detected participants' face images. Namely, we utilized the extracted face embeddings, and calculated the Euclidean distance between all pairs of detected faces. We consider two face images to be images of the same user if the Euclidean distance between their vector representation is smaller than an empirically chosen threshold.[21]

Using the cross-referenced data, we constructed a social graph $G := < V, E >$ of the participants, where V is the set of video conference users, and E is a set of links between users that according to the collage image dataset participants in a meeting together.

In order to demonstrate that video conference meeting participants are susceptible to information leakage attacks, we tested whether it is possible to match video conference users with their online social networks' profiles by using the participants' usernames and facial images. We randomly selected video conference images from which we extracted 90 participants' faces with full names. For each participant, we manually searched for his/her username on social media sites and compared them to the username that was extracted from the video conference images. Whenever we found a social network profile with a matching username we downloaded the profile images of the profiles that had a visual resemblance to the video conference participant. Additionally, to validate the profiles we searched for the other participants in the friend/follower lists when the data was available. After collecting the social network images, we performed face detection and face encoding similar to those performed for the video conference collage images. Finally, we used these face encoding representations as a small face recognition database for evaluating profile linkage.[22]

Additionally, we inspected how the constructed network will be influenced if more users will take some measures to protect their privacy. We have simulated a scenario where we tested the number of edges the network will have in a case when we would not be able to cross reference $N\%$ of the participants. We run the simulation 100 times with ten different N values with an interval of 10 from 10% to 100%.

D. Linkage by Background

We investigated whether it is possible to use a user's unique background or unique geographical location to link the same user to participate in different meetings. To show that such linkage is possible, we first trained an object detection model using Faster region-based CNN (RCNN) to cut the collage image into blocks, where each block contains one user along with his/her background. We manually labeled objects from 56 collage images containing a total of 423 objects. The trained classifier achieved a mean average precision[23] (mAP) of 0.85, and a mean average recall (mAR) of 0.64 on a test

[18]The list consists of the following words: "help," "view," "meeting," "edit," "participant," "speaker," "security," "screen," "record," "video," "stop," "share," "manage," "view," "exit," "full," "chat," "end," "reactions," "recording," "mute," "zoom," "participants," "from," "recording," "window," "search," "invite," "leave," "unmute," "option," "delete," "raise," "hand," "new," "type," "message," and "here."

[19]The selected threshold of 10 was chosen, by manually testing and evaluating several threshold values.

[20]We used a strict measurement where even if only one character is different between the actual username and detect one it will not be considered as correct detection.

[21]To determine that two face images are of the same person, we empirically chose a distance threshold between the vector representation of two faces to be less or equal to 0.3, which is lower than the default distance value of 0.6 used by the *Face Recognition* python library [39]. We chose a relatively strict threshold since many of the faces extracted from Zoom collage images suffer from low resolution, thus resulting in many false matches of pairs of faces.

[22]We used the default values for face verification (threshold of 0.6).

[23]Pascal VOC metric.

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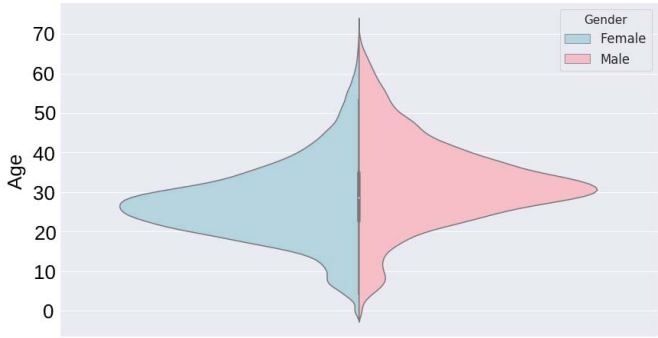

Fig. 3. Zoom users: gender and age distribution.

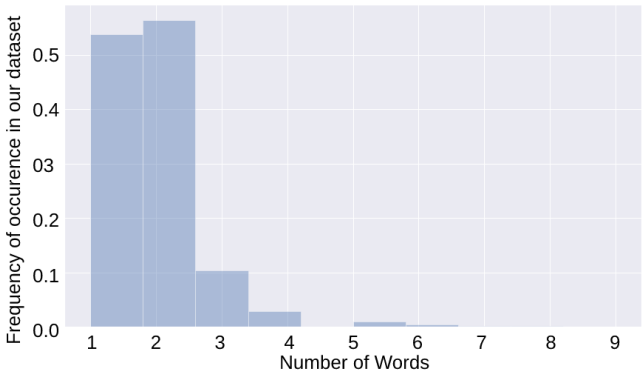

Fig. 4. Distribution of the number of words composing a zoom username.

set of nine images containing a total of 66 objects. Next, we used a pretrained ResNet-50 model to extract from each image feature representation from the model's last layer, and calculated the Euclidean distance between each two block images. To evaluate the method we inspected the first 100 pairs with the smallest Euclidean distance.

IV. Results

To analyze privacy threats in video conference collage images, we collected 15 783 Zoom collage images that were publicly posted on Twitter and Instagram. We explored the dataset of the collected Zoom collages and found an average of 9.04 participants in a video conference meeting collage. From the crawled Zoom collage images, we extracted a total of 142 001 face images for analysis. From an age perspective (see Fig. 3) the population of the posted collage images on social media is relatively homogeneous. The empirically computed average estimated age of the participants is about 29 and the median is about 28. By examining age distributions, we observed that 87.49% of the participants are adults, 6.18% children, 6.23% adolescents, and only 0.09% of participants are older adults. Additionally, the gender prediction algorithms identified 29 048 and 50 221 males and females, respectively.

We were interested in how discrete users are about their real-world identities, as reflected in the choice of their usernames. We thus extracted usernames from the Zoom image collages, and found 85 616 different usernames, out of which 48 818 (57% of the usernames) consist of more than a single word (see Fig. 4), suggesting possibly more disclosed information about the user. By manually inspecting these multiword usernames we observed that many of these words represent distinct names

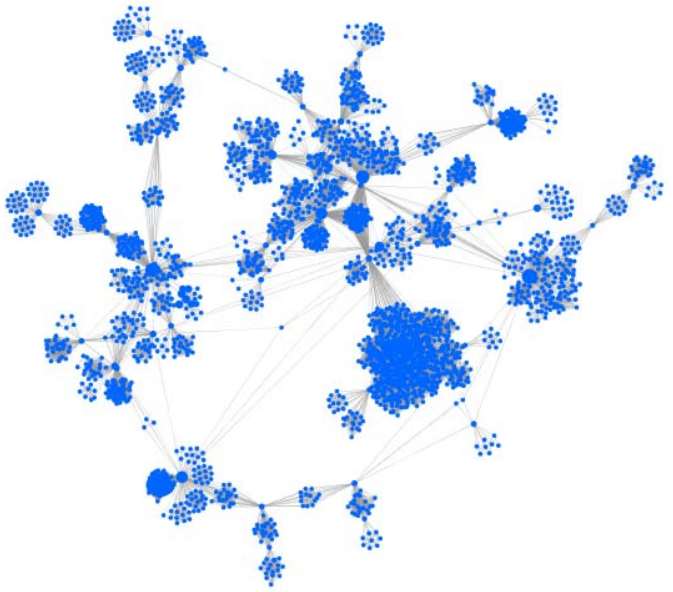

Fig. 5. Largest component observed in the social network of zoom users in our dataset. The network was constructed by matching meeting participants' faces and usernames across meetings as explained in methods. Each node in the network represents a Zoom meeting participant, and each edge between users in the network represents that both users participated in at least in one video conference meeting.

that can be utilized to match a user's social network profile. In fact, 2522 of these usernames appeared in several collage images from different meetings.

Our analysis of the participant's facial images, we identified 1153 faces that likely appeared in several different Zoom meeting images. Using the cross-referenced images, we constructed a large-scale social network of Zoom users with 16 842 nodes (participants) and 197 765 edges (participation in joint meetings). Each node in the network represents a single meeting participant, and each edge between a pair of participants in the network represents that they jointly took part in at least one video conference meeting. The network is composed of 345 distinct connected components. On average, each separate component consisted of 48.8 participant nodes and 573.2 joint meeting edges. The largest component consisted of 3066 nodes and 55 035 edges (see Fig. 5).

To evaluate the profile linkage with social networks 90 random participants were selected. For each of the 90 participants, we searched for a matching network profile, possibly depicting the same person. We found 63 such matches. Additionally, through these profiles' friends/followers lists, we found nine other matches of profiles with partial names that look similar to the original video conference participant. Using face verification algorithms, we validated 49 social network profiles out of these 63 profiles, and five profiles out of the nine with partial names. We also observed networks where all participants were coworkers. Furthermore, we found multiple examples of users that are more privacy aware than others, as evidenced by their use of smiley emojis over the meeting participants' faces in order to protect their privacy. We also simulated the influence of additional participants taking measures to protect their privacy (see Fig. 6). We find that even if only 20% of the users will take some measures to protect only themselves, it will reduce, on average 36%, of the links in the network.

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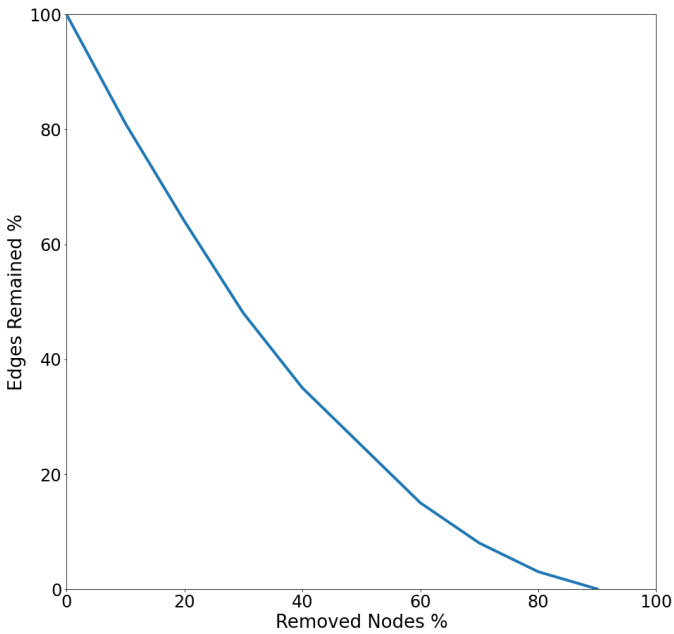

Fig. 6. Reduction in the number of edges in the network as a function of removed nodes.

Analyzing the video conference participants' image backgrounds, we discovered that 96 out of the 100 inspected pairs had the same background. Furthermore, out of the 96 pairs, 50 consisted of completely different images, while 46 were basically the same image with minor modifications, such as different filters, and changed resolutions.

V. Discussion

Our results point to several important findings as follows.

1) *MP Got Way Harder:* The results emphasize how publishing a single video conferencing collage image affects multiple people's privacy. In many online social networks, user can only protect their privacy in MP scenarios by controlling in which photographs they are tagged or by reporting a privacy violation to the social network operator. Unlike regular multiparty images, video conference images have the username embedded in the image. In other words, even if a user has disabled image tagging in their social network privacy settings, his/her name will still be written under the image. The current approaches for dealing with MP are mainly focused on MP conflicts (how to decide if the image should be published) [46]. In video conferencing, collage images can contain dozens of participants, which makes the decision even harder. The most practical solution is similar to the fine-grained approach [46]. We recommend blurring all the faces and names of participants and unblurring them only with the explicit consent of the participant.
2) *Collecting Private Data From Publicly Available Images Is Easy:* We demonstrate that by applying Data Science tools and machine learning algorithms, it is relatively easy to collect personal data from an abundance of Zoom collage images publicly posted on the Web. While almost any type of image might leak some data, the likelihood of data leakage in video conference images is higher. Almost, all video conference images leak personal data through their unique structure that contains data that is not available in regular images. Using Instagram and Twitter crawlers, we collected over 15 000 collage images taken during Zoom video conference meetings. Using face detection algorithms on those Zoom collages, we extracted a dataset of over 140 000 faces and over 85 000 distinct usernames. These results indicate that many images taken from video conference meetings are already publicly available online, and relatively easy to obtain. Moreover, the collected face images and usernames can further be utilized to obtain personal information about meeting participants, including their facial characteristics, age, gender, usernames, and sometimes even full names. This type of extracted data can vastly and easily jeopardize people's security and privacy both in the online and real world. Furthermore, in this study, in order to make this point, we generated our dataset by collecting publicly available data only. However, malicious actors (such as Zoombombers) can potentially actively harvest data from video conference meetings and construct more precise facial datasets of video conference participants' data without their consent and knowledge. Users can take preventive measures to protect themselves making data harvesting harder. For instance, they can change their privacy setting on social media making their post available only to friends. In such case, to collect data and perform a similar analysis, an attacker will have to create social bots that will try to create friendship connections with users on social media.
3) *Privacy Risks Affect All Age Groups, Even Young Children and Older Adults:* The collected dataset provides a demographic glimpse into the world of video conference meetings. By analyzing the age and gender features of over 140 000 participants (see Fig. 3), we observed that various population segments widely use video conferencing. While the vast majority (87.49%) of meeting participants were adults, as might be expected considering that due to COVID-19 many companies have been encouraging employees to work from home [47], the age span is in fact from children to older adults. This result highlights the risk that privacy and security issues may affect not only adults but also more vulnerable segments of the population, such as young children and older adults.
4) *Privacy Risks Affect Young Children:* According to our dataset about 12% of the people on video conference images on social media are minors. While this is not a large percentage of the users, they should still be considered a group that needs special attention. Young children are a particular case regarding privacy, as they are considered the most vulnerable age group online. All major social networks have an age restriction of 13 in the U.S. and 16 in Europe. Hence, children and teenagers can not be tagged even if their photographs are uploaded online. However, unlike regular photographs, video conference collage images have names embedded near each person, effectively serving as a tag.

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Additionally, many laws limit the publication of information and photographs of minors. For instance, the Family Educational Rights and Privacy Act (FERPA), is a U.S. federal law "that protects the privacy of student education records" [48]. In other words, in order to publish students' information online, including photographs, explicit parental consent is required [49], [50]. The Children's Online Privacy Protection Act (COPPA) is another U.S. law that deals with minor privacy online. COPPA protects the privacy of children under 13 online and limits children's data collection by online websites without parental consent [51]. We demonstrated how easily private information, such as name, gender, and age, could be collected from video conference photographs that contradict these legal requirements. It is also reasonable to assume that photographs that educators upload may contain additional textual metadata such as school name and class. In other words, even though the children do not have online social network accounts, their private information is published, shared, and stored on the Internet. The issue is becoming more crucial since a recent survey conducted by Google shows that most parents are worried about their children's privacy while attending school online [52]. In addition, Kaspersky [53] found that most parents are worried about their child's online safety but are not doing anything about it. Thus, even though parents may appear to be unaware due to their inactivity, they still view these technologies as a privacy risk.

5) *Privacy and Security of Organizations at Risk:* We showed that cross-referencing facial image data with social network data may put participants at additional privacy risks they may not be aware of. There are indications that many users are unaware of the standard privacy issues in social networks [54]; video conferencing presents a new type of privacy issue that may be even more challenging to notice. In fact, as one example, we explored a meeting that contained a group of adults. By performing an in-depth analysis of these images and comparing the participants' usernames and facial images with data available from online social networks, we were able to conclude that the meeting participants worked in the same company and that we could additionally infer social links among the meeting participants. This illustrates that not only individuals' privacy is at risk from data exposed in video conference meetings, but also the privacy and security of organizations. For example, by inspecting the social network of participants in an organization, it is possible to expose various private details regarding the organization itself [55]. Thus, an attacker can potentially utilize the organization's social network in order to attack the organization [56]. Unlike standard methods for information collected from online social networks using video conferencing data, we can establish that two employees know each other and probably work on the same project and maybe even on the same team. This type of information is not publicly available on many online social networks, such as LinkedIn. Another issue is that publishing video conference images by companies or businesses on social media can be considered a violation of the General Data Protection Regulation (GDPR). GDPR defines "personal data" as "any information relating to an identified or identifiable natural person ("data subject"); an identifiable natural person can be identified, directly or indirectly, in particular by reference to an identifier such as a name, an identification number, location data, an online identifier or to one or more factors specific to the physical, physiological, genetic, mental, economic, cultural or social identity of that natural person" [57]. In other words, publishing video conference images of a customer or an employee can result in a GDPR violation since the participants can be identified by name or by their face.

6) *Information From Different Meetings Could Be Aggregated to Uncover Social Networks and Be Used for Linkage Attacks:* We demonstrated that by using meeting participants' facial characteristics and usernames, it is possible to identify users that appear in several video conference meetings, thus collecting and aggregating their data. For example, as depicted in Fig. 5, an individual's social network could be constructed by combining data from several meetings he or she participated in. Moreover, we demonstrate that it is possible to use data collected from video conference meetings along with linked data collected in other video meetings with other data sources, such as online social networks, in order to perform a linkage attack on target individuals. This can jeopardize the target individual's privacy by using different meetings to discover different types of connections. It is possible, for instance, to aggregate information about a person's job, personal, and political connections. We observed that some individuals are evidently aware of the dangers of publishing video-conferencing images on social media, as can be inferred from meeting images posted with blurred names or faces of some of the participants, or similarly, from posted collage images in which large emojis were pasted on the participants' faces to protect their privacy.

7) *Unique Backgrounds Could be Used Against You:* We note that some meeting participants were more successful than others in protecting their privacy. Such privacy protection practices included using generic usernames and replacing their faces with smiley images. Interestingly, even for such privacy conscious individuals, we realized that it is possible to identify the same individuals across different meetings by using other means. We demonstrated that by clustering similar backgrounds, we could identify that same user by unique backgrounds, another potential privacy risk that users may be unaware of. We speculate that there is potential to discover additional private information by analyzing the video conference participants' background images. We plan to explore this assumption further in a future study by utilizing advanced machine learning methods to analyze unique background images.

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VI. Recommendations for Privacy Protection

We demonstrated throughout this article that video conference users face various security and privacy threats. By deploying some protective measures, participants can reduce the amount of data that is available to malicious actors, thereby protecting not only themselves but other participants as well. We suggest several easy-to-apply methods which can assist video conference participants in improving their security and privacy as follows.

1) *Use Voice-Only Whenever Possible and Avoid Uploading Virtual Meeting Photographs and Videos Onto Social Media:* This article shows that images taken from video conference meetings can jeopardize individual security and privacy in multiple ways. It is sufficient for a single collage image to be taken by one of the participants to jeopardize the privacy of all associated meeting participants. Therefore, whenever possible, we suggest that video conference users avoid video streaming whenever possible and use voice only in situations that allow it. Avoid uploading videos and collage photographs of video conference meetings onto social media. That being said, a recent study [20] demonstrated that even when the conversation is muted, it still transmits data. We recommend that users who require a high level of privacy use a microphone with a hardware turn on/off button to ensure that audio data is not transmitted.
2) *Use Generic Pseudonyms for Video Conferencing:* This article demonstrates, using a full name or a unique username for video conferencing can easily expose an individual to attacks. Moreover, using a generic name, such as "iPad" or "iPhone" makes it more challenging to cross-reference a user's identity with other datasets. Therefore, we strongly recommend that video conference participants not use their real names and advise choosing a generic pseudonym instead.
3) *Use Generic Background Images as Virtual Backgrounds:* While streaming video, having a virtual background is highly recommended. Using a real background can compromise a user's security and privacy. For example, items in the background of an individual's room can reveal his/her identity and even geographical location. Additionally, not using a virtual background or choosing a unique background image may help malicious users in fingerprinting a user account across several meetings. However, even a generic background may not be sufficient for a live video conference [19], [21]. Reflection from eyeglasses and artifacts from the background generation algorithm may leak private data [21]. Currently, the most effective solution to these problems is using better matting algorithms, but this is up to the service operators. Until a better algorithm is implemented in the video conferencing platform, the end-user should continue to use a generic background. Also, choosing a location without distinctive elements is recommended to reduce the threat. Users wearing eyeglasses should consider what kind of data they browse during video conference meetings.
4) *Use Anti-Facial Recognition Accessories:* In recent years, some companies have started selling anti-facial recognition clothing and makeup to protect the users' privacy. Some accessories can be homemade. For example, Sharif et al. [58] presented a method using a home printer to create a skin for glasses that disrupt facial recognition algorithms. Moreover, we observed that in many cases, we could not to detect the faces of people wearing even a basic party mask. Another indication that a simple mask may interfere with facial identification is the problems facing users that try to unlock their smartphones while wearing surgical masks. Recently, Apple updated its Face ID software to enable user identification while wearing surgical masks. Simple masks will not provide complete protection but should disrupt most generic face verification algorithms. We also observed that some users replaced their faces with smiley images to hide their identities, another reasonable option to consider for hiding user identity.
5) *Organizations Should Inform Employees on Video Conference Privacy Risks:* Many organizations, especially during the pandemic, encourage their employees to work from home and conduct meetings using video conference applications. This results in a new set of security and privacy threats to organizations that not all organizations are aware of. Here we demonstrated that sharing data from video conference meetings can compromise organizations' security and privacy. Therefore, we recommend that organizations update their policies on what employees can share and what should not be disclosed while using video conference applications.
6) *Monitor Children's Video Conferencing Activity:* Children are active participants in video conference meetings. While a video conference may seem a safe and harmless place to be, they are at risk of numerous threats ranging from malware to cyberbullying and Zoombombing. Video conferencing is not different from regular online activity, and parents should be actively involved in monitoring their children's activity in the same way. To reduce risks, parents should explain to their kids not to accept a request to join a meeting from someone they do not know in real life. In addition, they should remove personal details from the child's account and adjust the privacy settings. The meeting organizer should use features such as the waiting room to prevent uninvited individuals from attending the meeting.
7) *Video Conference Operators Should Add and Support Privacy Mode:* In the past couple of years, researchers presented methods that can disrupt facial recognition. For instance, Wilber et al. [59] showed that adding Gaussian noise to an image can disrupt facial recognition while keeping the face still recognizable to humans. Even without the help of video conference operators in some applications, users may opt to use filters in order to hide their appearance and avoid automatic facial recognition. Video conferencing applications should include a notification being sent to all participants when someone takes a screenshot, similar to some instant massaging applications.

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8) *Tidy Up Social Networks Privacy Settings:* Social networks can be easily used to mine data from video conference meetings. While it is unlikely that people will stop uploading such images, they can tighten their privacy settings. We suggest making posts visible only to friends. This simple change makes it much more difficult for malicious actors to collect data.

VII. Summary and Conclusion

The COVID-19 pandemic has drastically and globally changed how people communicate, immensely increasing communication via video conferencing. As a result, novel privacy and security issues have emerged, endangering hundreds of millions of video conference participants. In this study, we dived into the privacy issue of video conference applications by analyzing over 15 000 video conference images of over 140 000 meeting participants. From these images, we extracted multiple private information features including gender, age, and real name. We demonstrated that this information could be utilized to uncover additional details about video conference participants.

We demonstrated that cross-referencing participants in multiple conference meetings collage images from public sources could uncover the participant's social network. Also, we show that even the user's background may be used to discover social relations. It subsequently may compromise his/her or other participants' privacy and social accounts. Specifically, this makes MP way harder, requiring better solutions to protect user's privacy. That said, we offer several recommendations in which users can protect themselves, for instance, by using generic usernames and backgrounds, wearing small masks, or using a filter. In the current global reality of social distancing, we must be sensitive to online privacy issues accompanying lifestyle changes as society is pushed toward a more virtual world.

Data availability

The data will be available upon request.

Acknowledgment

The authors thank Valfredo Macedo Veiga Junior (Valf) for designing the infographic illustration.

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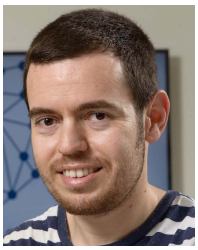

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